

Meta WhatsApp Cloud API & Gateway Integration Guide

Comprehensive Walkthrough for Setup, Developer Configuration & REST API Gateway Deployment

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1. Meta Developer Account Registration

To use the WhatsApp Cloud API, you first need to register as a Meta Developer:

1. Visit developers.facebook.com.
2. Click **Get Started** (or **Log In**) in the upper-right corner. Log in with your standard Facebook credentials.
3. Complete the registration flow:
 - Accept the Meta Developer Terms and Conditions.
 - Verify your phone number via SMS OTP code.
 - Choose your developer role (select **Developer** or **Product Manager**).
4. Click **Complete Registration**.

IMPORTANT

Two-Factor Authentication (2FA) is strictly required by Meta for all developer profiles accessing Business tools like WhatsApp. Make sure your Facebook profile has 2FA enabled in your Security settings.

2. Creating a Meta App

Next, you need to create an application wrapper in Meta to access the APIs:

1. Navigate to the **Meta App Dashboard** (developers.facebook.com/apps).
2. Click the green **Create App** button.
3. Select **Other** for the use case and click **Next**.
4. Choose **Business** as the app type.

NOTE

The **Business** app type is highly recommended because it allows you to connect the application directly to a Meta Business Portfolio (formerly Business Manager), which is required for WhatsApp.

1. Provide the App Details:
 - **App Display Name:** E.g., [Clinic WhatsApp Gateway](#)
 - **App Contact Email:** Your active email.

- **Business Account:** Select your existing Meta Business Portfolio if you have one. If you do not have one, leave it as *No Business Manager portfolio selected* (Meta will automatically create a temporary one for you).

2. Click **Create App** and input your Facebook account password for confirmation.

3. Adding the WhatsApp Product

Once your application is created, you will be redirected to the App Dashboard:

1. Scroll down to the **Add products to your app** section.
2. Locate **WhatsApp** and click the **Set up** button.
3. Choose or create a Meta Business Portfolio to link with the App, then click **Continue**.
4. Accept the WhatsApp Business Terms of Service.

4. Acquiring Test Credentials & Sending First Test Message

Once WhatsApp setup is complete, you will be taken to the WhatsApp **API Setup** page:

Step 4.1: Copy Sandbox Credentials

Look at the **Send and receive messages** dashboard. Under **Step 1: Select phone numbers**, copy the following credentials:

1. **Temporary Access Token:** Valid for **24 hours**.
2. **Phone Number ID:** A numeric ID (e.g., `105347239243789`). This represents your sandbox phone number.
3. **WhatsApp Business Account ID:** A numeric ID (e.g., `101037894234589`).

Step 4.2: Add a Recipient Test Number

Because your app is in Sandbox mode, you can only send messages to pre-registered test numbers:

1. In the **To** field dropdown under step 1, select **Manage phone number list**.
2. Select your country code and enter your personal WhatsApp phone number (e.g., `919876543210`).
3. Complete verification by typing the OTP code sent to your WhatsApp.

Step 4.3: Trigger a Sandbox Message

1. Click the **Send message** button on the dashboard.
2. Check your phone. You should receive a template message reading *"Welcome to the Meta WhatsApp Cloud API..."*.

Section 5: Setting Up a Permanent Access Token (System User Token) — [Skipped / Not Implemented]

Section 6: Adding a Live/Production Phone Number — [Skipped / Not Implemented]

7. Configuring the WhatsApp Gateway Project

This repository contains the full WhatsApp Gateway code split into a Node.js `backend` and a Next.js `frontend`.

Step 7.1: Configure Backend Environment Variables

1. Navigate to the `backend` directory.
2. Copy the `.env.example` file to a new file named `.env`:

```
cp .env.example .env
```

3. Open the `.env` file and configure your values:

```
PORT=5000
MONGODB_URI=mongodb://127.0.0.1:27017/whatsapp-gateway
JWT_SECRET=super_secret_jwt_key_for_whatsapp_gateway_dashboard
GATEWAY_API_KEY=your_hms_gateway_secret_api_key_2026
NODE_ENV=development
```

Step 7.2: Install and Start the Backend

1. Open a terminal in the `backend` directory:

```
npm install
```

2. Run the seeding script to create the default dashboard login account:

```
node scripts/seedAdmin.js
```

This seeds an administrator account with:

- **Username:** `admin`
- **Password:** `adminPassword123`

3. Run the development server:

```
npm run dev
```

The backend will boot up at `http://localhost:5000`.

Step 7.3: Install and Start the Frontend Dashboard

1. Open a terminal in the `frontend` directory:

```
npm install
```

2. Start the development server:

```
npm run dev
```

The dashboard will boot up at `http://localhost:3000`.

8. Registering a Clinic & Sending Messages via the Gateway API

Step 8.1: Add the Clinic Configuration in Dashboard

1. Open your browser and go to `http://localhost:3000`.
2. Log in using the seeded credentials:
 - **Username:** `admin`
 - **Password:** `adminPassword123`
3. Click on the **Clinics** tab and select **Add Clinic**.
4. Fill in the credentials obtained from the Meta Developer Dashboard:
 - **Name:** E.g., `Star Dental Clinic`

- **Code:** E.g., `star-dental` (Unique ID used in API requests)
- **Sender Mobile:** The verified phone number from Meta (e.g., `919876543210`)
- **Phone Number ID:** Paste the Phone Number ID from Step 4.
- **WhatsApp Business Account ID:** Paste the Business Account ID from Step 4.
- **Access Token:** Paste your Access Token.

5. Click **Save**.

6. Find the newly created Clinic in the dashboard list and click on it to retrieve its auto-generated **API Key** (e.g. `hms_key_a8d7a12bc9f34...`).

Step 8.2: Send Messages via the Public REST API

Your Clinic Software/HMS can now trigger WhatsApp messages by making standard REST API requests.

Endpoint: **POST** `http://localhost:5000/api/send-message`

Headers:

- **Content-Type:** `application/json`
- **x-api-key:** `hms_key_a8d7a12bc9f34...` (Replace with your Clinic's unique API Key)

Request Body:

```
{
  "clinicId": "star-dental",
  "mobile": "919876543210",
  "message": "Hello! Your appointment at Star Dental is confirmed for tomorrow at 11:00 AM."
}
```

Curl Example:

```
curl -X POST http://localhost:5000/api/send-message -H "Content-Type: application/json" -H "x-api-key: hms_key_a8d7a12bc9f34..." -d '{
  "clinicId": "star-dental",
  "mobile": "919876543210",
  "message": "Hello! Your appointment at Star Dental is confirmed for tomorrow at 11:00 AM."
}'
```

9. Webhooks Setup (Optional for Status Tracking)

To receive real-time updates when a customer reads a message or sends a reply, you can subscribe to Meta's Webhooks:

1. In the Meta Developer Dashboard, expand **WhatsApp** and click **Configuration**.
2. Under the **Webhook** section, click **Edit**.
3. Set the fields:
 - **Callback URL:** The public endpoint of your backend (e.g., `https://your-domain.com/api/webhook`).
*For local development testing, you can use **ngrok** to create a secure tunnel: `ngrok http 5000`
 - **Verify Token:** A secret string of your choosing (e.g., `my_private_webhook_token_123`) which you configure in your backend configuration.
4. Click **Verify and Save**.
5. Under the **Webhook fields** dashboard section, click **Manage** and subscribe to:
 - `messages` (triggers when a text message is received or updated to *delivered/read* status).